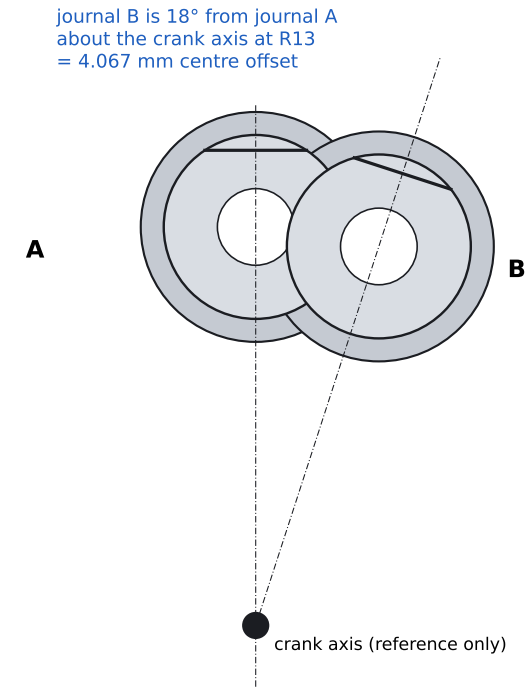
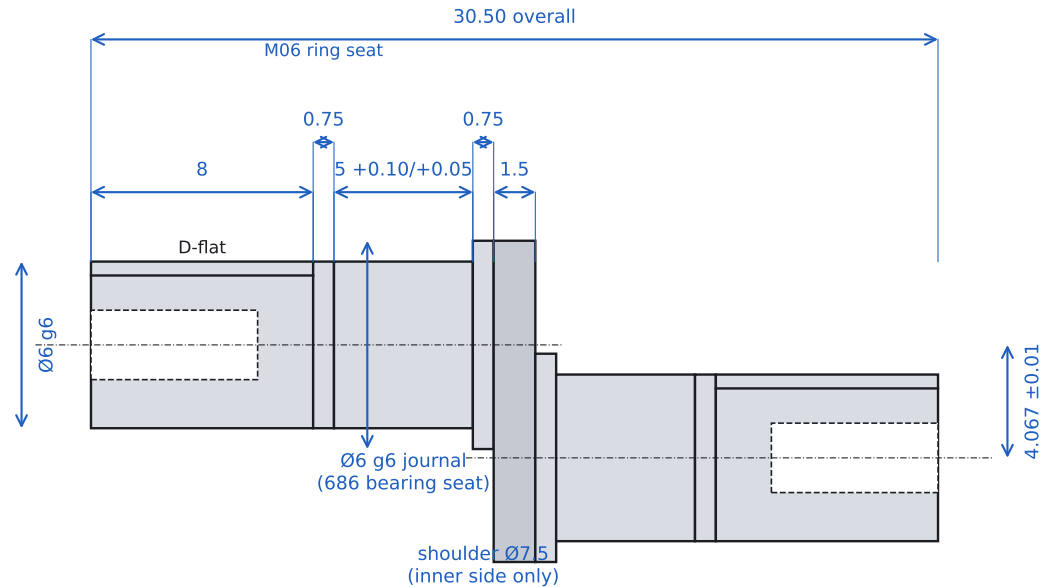


M01 Split crankpin

Material: Stainless 303 / 416 (or 1144 steel, black oxide) Qty: 5 per engine / 250 for 50 units Units: mm V10 display engine (original design)

SIDE VIEW (end A = front journal)

END VIEW from end A

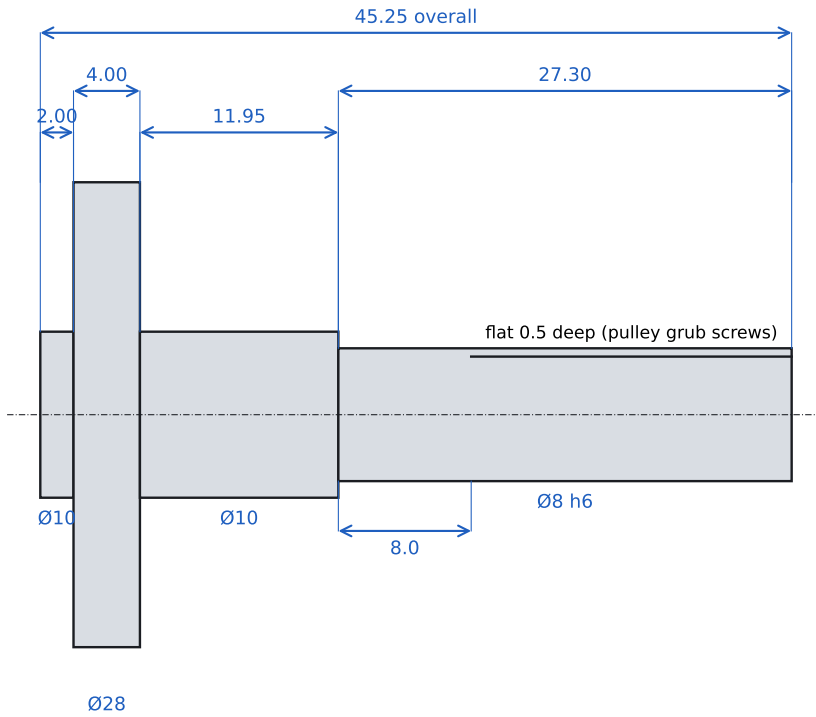


- Ø6 g6 (-0.004/-0.012) from each end up to its shoulder (journal, ring seat and end are ONE diameter), Ra 0.4 on the journals.
- Shoulders Ø7.5 ONLY on the inner side of each journal: the 686 bearing and the M06 ring slide on over the end.
- Journal centre offset 4.067 ±0.01 at 18° about the crank axis (R13). All other dims ±0.05.
- D-flats 0.5 deep, each facing radially AWAY from the crank axis (see end view). Flats key the pin into the printed webs.
- Both ends tapped M3 x 6 deep (tap drill Ø2.5). Break all edges 0.2 (0.3 x 45° lead chamfer on both ends). Shoulder faces square within 0.02.
- Journal B side mirrors journal A (same lengths). The side view is drawn in the plane of the offset.
- The part is symmetric end-for-end: it cannot be fitted the wrong way round.
- Suggested process: turn Ø12 bar, journal A + shoulder; re-chuck on 4.07 offset (or mill-turn) for journal B; mill the stadium web and flats.

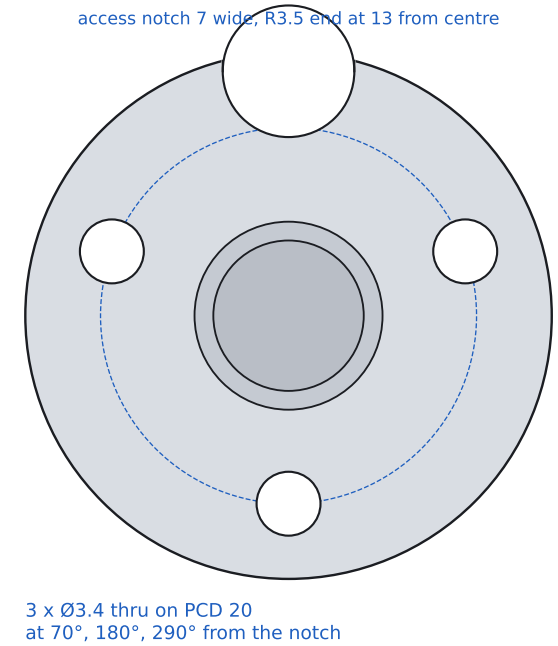
M02 Main shaft (front and rear identical)

Material: Stainless 303 (or 1045, black oxide) Qty: 2 per engine / 100 for 50 units Units: mm V10 display engine (original design)

SIDE VIEW



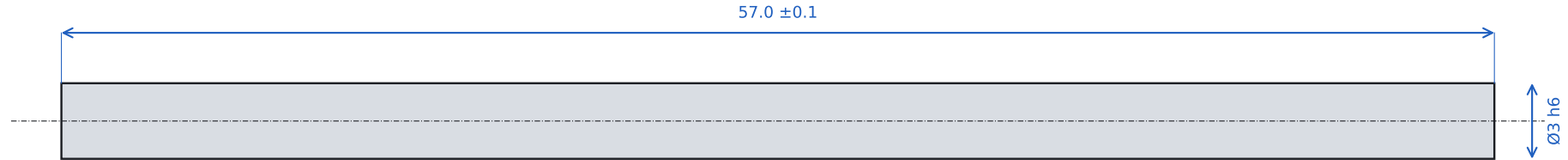
FLANGE END VIEW



- Ø8 journal h6 (0/-0.009) for the 608 bearing; Ra 0.8. Shoulder face square to axis within 0.02 (it locates the bearing inner ring).
- Flange face square to axis within 0.02; spigot Ø10 -0.02/-0.05 (locates in the printed end web). Other dims ± 0.05 .
- Flange holes are clearance holes; the UNEVEN bolt pattern (110/110/140° spacing) plus the notch makes the shaft fit the web one way only.
- The flat is only needed on the FRONT shaft (drive pulley). Machining it on both keeps one part number.
- Alternative if machining capacity is short: 8 mm h6 ground rod + commercial 8 mm rigid flange coupling (lower precision; not recommended for production).

M03 Piston guide rail

Material: 3 mm ground stainless rod (h6) Qty: 10 per engine / 500 for 50 units

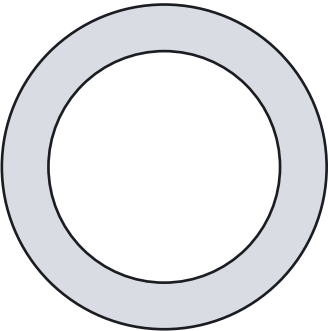
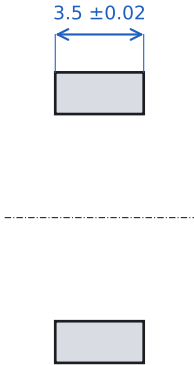


- Cut from 3 mm h6 ground stainless linear rod (not drill rod - needs corrosion resistance and a polished surface).
- Both ends: $0.3 \times 45^\circ$ chamfer, deburred so they do not scratch the bronze bushings. Length ± 0.1 .
- The rail is the ONLY thing the piston slides on (two oil-impregnated bronze bushings). Keep it clean; one drop of light oil at assembly.

M04 Pulley spacer

Material: stainless 303 Qty: 1 per engine / 50 for 50 units

SECTION



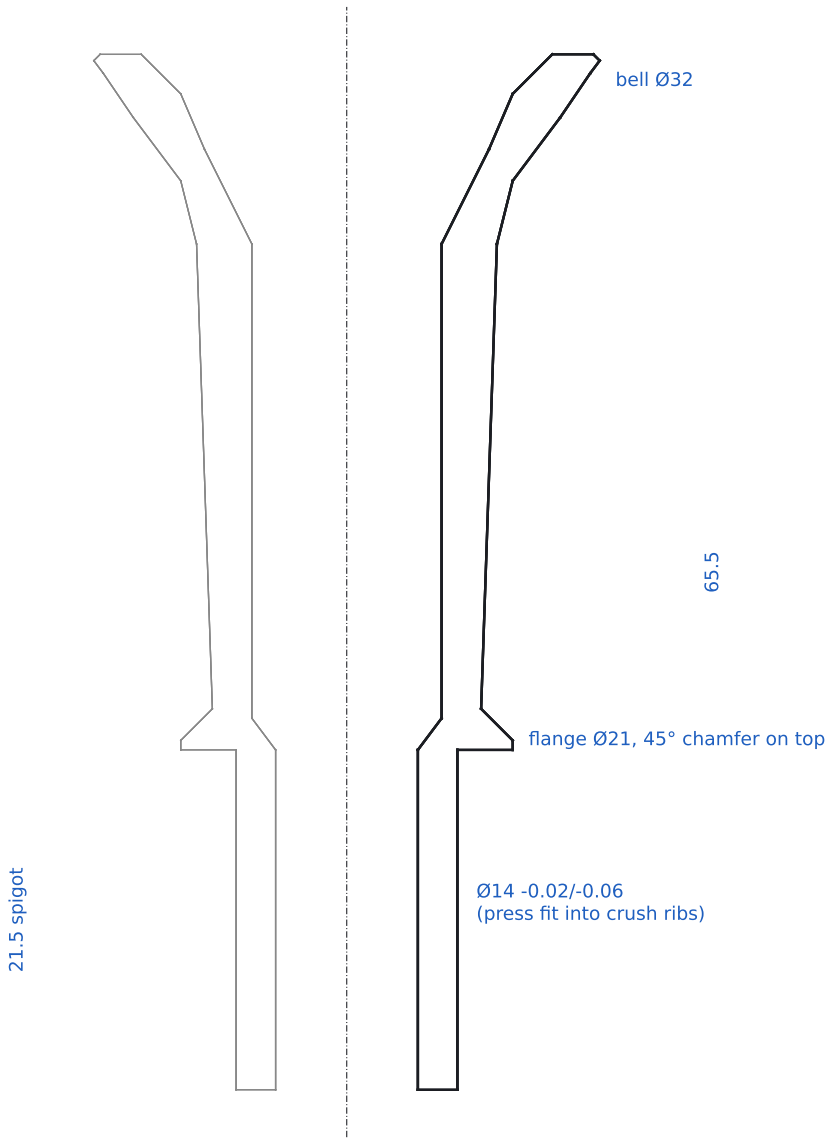
OD 11.5 -0.1 (must stay inside the 608 inner ring) ID 8.2 +0.05

- Sits on the front main shaft between the 608 inner ring and the 60T pulley hub; clamps the crank's axial position.
- Faces flat and parallel within 0.02. Break edges 0.2. Can be parted off Ø12 bar in the same set-up as M02.

M05 Intake trumpet (optional machined upgrade of printed part 13)

Material: Aluminium 6061-T6, clear or black anodised Qty: 10 per engine / 500 for 50 units Units: mm V10 display engine (original design)

HALF SECTION (right) / outline (left), mm



- Turned from Ø35 bar. Profile exactly as the STEP file stl/M05_intake_trumpet_machined.step (this drawing is a guide).
- Spigot Ø14 -0.02/-0.06: it goes through the throttle body and is held by the crush ribs in the head port.
- Bore and bell inside: Ra 1.6 or better, polished. Outside: Ra 0.8, bead-blast + anodise for the premium finish.
- Break all edges 0.2. No other tolerances are critical (general ± 0.1).

M06 Bearing spacer ring

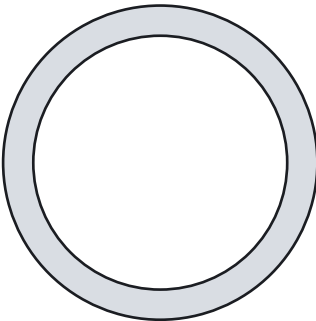
Material: stainless 303 (or hardened steel shim stock) Qty: 10 per engine / 500 for 50 units

SECTION (scale 6:1)





FACE (scale 1:1 grid)



OD 7.5 -0.05/-0.10 ID 6 +0.03/+0.08

- Slides onto each crankpin end after the 686 bearing; clamps the bearing inner ring against the pin's shoulder.
- OD must stay inside the 686 inner ring face (≤ 7.5) so it never touches the bearing's outer ring or shield.
- Faces flat and parallel within 0.01, deburred both sides (a burr would tilt the bearing). Parted off $\varnothing 8$ tube or bar.